SKYINSIGHT

Deep Learning

Computer Vision Module

3 Vision Models · Enhanced Pipeline · Transfer Learning

Objectives

01

Automate Visual QC

Replace manual inspection with AI-powered image classification across three airline operation domains — cabin, crowds, and baggage.

02

Robust Deep Learning Pipeline

Build a reproducible, state-of-the-art training pipeline with transfer learning, advanced augmentation, and rigorous evaluation protocols.

03

Explainable & Production-Ready

Deliver interpretable models via Grad-CAM visualizations and TTA, ready for integration into the SkyInsight BI platform.

Data Cleaning & Preprocessing

Dataset Structure

Cabin Cleanliness

Classes: Clean / Dirty

Split: Train · Val · Test

Crowd Analytics

Classes: Low / High Crowd

Split: Train · Val · Test

Baggage Handling QC

Classes: Damaged / Good

Split: Train · Val · Test

Preprocessing Pipeline

1. **Resize & Normalize**

   All images → 224×224px · Rescale ÷255 · Backbone-specific normalization

2. **Data Augmentation**

   Rotation · Shift · Zoom · Flip · Brightness · Channel shift · Reflect fill

3. **Mixup Augmentation**

   Beta(0.2, 0.2) mixing of pairs — improves generalization & reduces overfit

4. **Class Balancing**

   Auto class-weight with label smoothing (ε=0.05) for imbalanced datasets

Implemented Models

Cabin Cleanliness

Detection

Backbone

EfficientNetV2S

Classes

Clean / Dirty

Epochs

20 head + 15 fine-tune

Loss

Focal Loss (γ=2.5)

Unfreeze top 40% · Dropout 0.45/0.30

Crowd Analytics

Classification

Backbone

EfficientNetV2S

Classes

Low / High Crowd

Epochs

22 head + 10 fine-tune

Loss

Focal Loss (γ=2.0)

Unfreeze top 25% · No H-flip

Baggage Handling

Quality Control

Backbone

DenseNet121

Classes

Damaged / Good

Epochs

16 head + 12 fine-tune

Loss

Categorical Cross-Entropy

Unfreeze top 30% · Dropout 0.40/0.30

Enhanced Architecture & Training

Model Architecture

Input 224×224×3

↓

Backbone (EfficientNetV2S / DenseNet121)

↓

GeM Pooling (p=3) + BatchNorm

↓

Residual FC Head (256→256→128)

↓

Softmax Output

Key Enhancements

Optimizer	AdamW + weight decay (1e-4)
LR Schedule	Cosine Annealing + Warmup (10%)
Augmentation	Mixup Beta(0.2) on every batch
Evaluation	TTA × 5 passes for robust preds
Interpretability	Grad-CAM per class visualization
Checkpointing	Best val-F1 (not val-loss)
Precision	Mixed float16 for GPU speedup

Project Summary

Objective

- Automate airline visual inspection with CV
- 3 domains: cabin · crowd · baggage
- Integrated into SkyInsight BI platform

Data Pipeline

- 224×224 resize · backbone normalization
- Rich augmentation (rotation, zoom, brightness…)
- Mixup + class weighting + label smoothing

Models

- Cabin & Crowd → EfficientNetV2S
- Baggage → DenseNet121
- GeM Pooling + Residual FC Head

Enhancements

- AdamW · Cosine Annealing · Warmup
- TTA (×5) + Grad-CAM interpretability
- Best F1 checkpointing · Mixed precision